Rigidity toward the polynomially bounded shift-similarity problem

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Status. Candidate substantial partial result, likely valid, subject to human review. The full polynomially bounded problem is not resolved. We prove two affirmative structural cases, a local Riesz-rigidity theorem, and a cluster–orthogonality obstruction which rules out the most direct use of the 2024 cyclic polynomially bounded construction as a counterexample.

1 Source problem

Let S be the unilateral shift on H^2 and

$$h_\lambda(z) = \frac{1}{1 - \overline{\lambda} z}, \qquad u_\lambda = (1 - |\lambda|^2)^{1/2} h_\lambda \quad (\lambda \in \mathbb{D}).$$

Thus $\|u_\lambda\| = 1$ and $S^* u_\lambda = \overline{\lambda} u_\lambda$. The source is M. F. Gamal', *On power bounded operators with holomorphic eigenvectors, II*, arXiv:1901.03883 [2]. It asks whether the following contraction criterion of Uchiyama [6] remains true for polynomially bounded operators.

Question. Suppose $T \in \mathcal{L}(\mathcal{H})$ is polynomially bounded and there is a quasiaffinity $X : \mathcal{H} \to H^2$ such that $XT = SX$ and

$$0 < c \leq \|X^* u_\lambda\| \leq C < \infty \qquad (\lambda \in \mathbb{D}). \tag{1}$$

Must T be similar to S?

The source says explicitly that the question remains open; the relevant passage on source p. 2 is reproduced below.

> quasiaffine transforms of unilateral shifts of finite multiplicity is generalized to power bounded operators. Also, in [G2] an example of a cyclic power bounded operator T_0 is given such that T_0 satisfies sufficient conditions on contractions to be similar to the unilateral shift S of multiplicity 1, but *there is no* contraction R such that $R \prec T$. This example is based on some example from [MT].
>
> In this paper, it is shown that the additional assumption $T \sim S$ on a power bounded operator T (with the requested norm-estimates of complete analytic family of eigenvectors of T^*) also *does not* imply the similarity $T \approx S$. The constructed operator T has the following property: there exist two invariant subspaces $\mathcal{M}_1$, $\mathcal{M}_2$ of T such that $T|_{\mathcal{M}_k} \approx S$ $(k = 1, 2)$ and $\mathcal{M}_1 \vee \mathcal{M}_2$ is the whole space on which T acts. The same property takes place for polynomially bounded operators that are quasiaffine transforms of S [G3]. A question whether the criterion for contractions to be similar to S can be generalized to polynomially bounded operators remains open.

The 2019 paper disproves the power-bounded version even after requiring $T \sim S$. A later paper [3] constructs cyclic polynomially bounded operators not similar to contractions and quasisimilar to S. The latter examples are an immediate candidate for settling the displayed question negatively, but they do not state the estimate (1). Section 6 proves that their direct shift extension cannot satisfy it.

2 Proof intuition

There are two competing geometries. Polynomial boundedness lets bounded analytic interpolation act diagonally on the eigenvectors X^*u_λ. On every interpolating sequence this forces those normalized eigenvectors to be a Riesz sequence. This is genuine rigidity, and it proves the desired conclusion whenever an independent structural theorem first makes T similar to a contraction.

The hypothesis still sees only ordinary eigenvectors, not generalized eigenvectors in long Jordan chains. The 2024 construction hides its failure of complete polynomial boundedness in large finite blocks. Within each block it places increasingly many distinct eigenvalues in a compact pseudohyperbolic region, while the corresponding adjoint eigenlines are orthogonal. A bounded map cannot send almost parallel normalized Hardy kernels to uniformly nonzero orthogonal vectors. This simple observation pinpoints why the strongest known candidate construction misses the source hypothesis and why its long-chain obstruction cannot be imported verbatim.

3 Two affirmative regimes

Theorem 1 (Completely polynomially bounded case)**.** *Under the hypotheses of the source question, if T is completely polynomially bounded, then T is similar to S.*

Proof. Paulsen's theorem [5] gives an invertible L for which $R = LTL^{-1}$ is a contraction. Set $X_0 = XL^{-1}$. Then

$$X_0 R = S X_0, \qquad X_0^* u_\lambda = L^{-*} X^* u_\lambda.$$

The map X_0 is a quasiaffinity, and

$$\frac{c}{\|L\|} \le \|X_0^* u_\lambda\| \le \|L^{-1}\| C \qquad (\lambda \in \mathbb{D}).$$

Uchiyama's contraction theorem [6] therefore gives $R \approx S$. Since $T \approx R$, the conclusion follows. □

This identifies a sharp necessary feature of any negative example: it must be polynomially bounded but not completely polynomially bounded.

Theorem 2 (Unilateral weighted shifts)**.** *Let $Te_n = w_n e_{n+1}$ be a unilateral weighted shift on $\ell^2(\mathbb{N}_0)$ with every $w_n \neq 0$. If T is power bounded and there is a nonzero bounded X satisfying $XT = SX$, then $T \approx S$. In particular the source question has an affirmative answer for unilateral weighted shifts, without using polynomial boundedness or* (1)*.*

Proof. Put $P_0 = 1$ and $P_n = w_0 \cdots w_{n-1}$ for $n \ge 1$. If $f = Xe_0$, then intertwining gives

$$Xe_n = P_n^{-1} S^n f.$$

Because X is nonzero, $f \neq 0$: if $Xe_0 = 0$, the displayed relation gives $X = 0$. Consequently

$$|P_n| \ge \frac{\|f\|}{\|X\|} > 0.$$

Power boundedness supplies the reverse uniform estimate $|P_n| = \|T^n e_0\| \le \sup_k \|T^k\|$. Hence the diagonal operator $Le_n = P_n e_n$ is bounded and invertible. The identity $TL = LS$ proves $T \approx S$. □

4 Interpolating-sequence rigidity

Call $\Lambda \subset \mathbb{D}$ an H^∞-interpolating sequence if every bounded scalar family on Λ is the restriction of some $f \in H^\infty$. Let K_Λ denote an interpolation constant.

Theorem 3 (Local Riesz rigidity)**.** *Assume T has polynomial bound M, $XT = SX$, and* (1)*. For every H^∞-interpolating sequence Λ, the family*

$$v_\lambda = \frac{X^* u_\lambda}{\|X^* u_\lambda\|}, \qquad \lambda \in \Lambda,$$

is a Riesz sequence. Its Riesz constants depend only on M and K_Λ. Moreover X^* is bounded below on $\operatorname{clos\,span}\{u_\lambda : \lambda \in \Lambda\}$*.*

Proof. Fix a finite $F \subset \Lambda$ and signs $\varepsilon_\lambda \in \{-1, 1\}$. For every $\eta > 0$ there is a polynomial p with

$$p(\lambda) = \varepsilon_\lambda \quad (\lambda \in F), \qquad \|p\|_\infty \le K_\Lambda + \eta. \tag{2}$$

Indeed, first interpolate by $f \in H^\infty$ with $\|f\|_\infty \le K_\Lambda$. Radial dilation puts f_r in the disk algebra. For r sufficiently close to one, a finite Lagrange correction makes its values on F exact with norm at most $K_\Lambda + \eta/2$. Uniform polynomial approximation followed by a second, arbitrarily small Lagrange correction proves (2).

Since

$$T^* X^* u_\lambda = \overline{\lambda} X^* u_\lambda,$$

$p(T)^*$ acts on v_λ, $\lambda \in F$, as the prescribed sign change. Its norm is at most $M(K_\Lambda + \eta)$. Letting $\eta \downarrow 0$ shows that every sign-change operator on the finite family has norm at most $K_0 = MK_\Lambda$. It is its own inverse. For scalars a_λ, averaging over independent Rademacher signs gives

$$K_0^{-2} \sum_{\lambda \in F} |a_\lambda|^2 \le \left\| \sum_{\lambda \in F} a_\lambda v_\lambda \right\|^2 \le K_0^2 \sum_{\lambda \in F} |a_\lambda|^2.$$

Thus the v_λ form a Riesz sequence.

The normalized Hardy kernels on an interpolating sequence are also a Riesz sequence [4]. Combining their upper Riesz estimate, the lower estimate just proved, and $\|X^* u_\lambda\| \ge c$ shows that, on every finite linear combination,

$$\left\| X^* \sum a_\lambda u_\lambda \right\| \ge c_\Lambda \left\| \sum a_\lambda u_\lambda \right\|$$

for some $c_\Lambda > 0$. Taking closures proves the last assertion. □

The theorem does not by itself prove that X^* is bounded below on all of H^2. A discrete interior interpolating sequence does not furnish a sampling Riesz basis for H^2, and clustered kernels require information about divided differences or generalized eigenvectors that is absent from (1).

5 The cluster–orthogonality obstruction

Let

$$\rho(\lambda, \mu) = \left| \frac{\lambda - \mu}{1 - \overline{\mu}\lambda} \right|$$

be the pseudohyperbolic distance.

Lemma 4 (Cluster–orthogonality). *Let $A : H^2 \to \mathcal{K}$ be bounded. Suppose that the vectors Au_λ, $\lambda \in \Lambda$, lie in mutually orthogonal one-dimensional subspaces and*

$$\|Au_\lambda\| \geq c > 0 \qquad (\lambda \in \Lambda).$$

Then Λ is uniformly pseudohyperbolically separated. Quantitatively, for distinct $\lambda, \mu \in \Lambda$,

$$1 - \sqrt{1 - \rho(\lambda, \mu)^2} \geq \frac{c^2}{\|A\|^2}. \tag{3}$$

Proof. The normalized Hardy kernels satisfy

$$|\langle u_\lambda, u_\mu \rangle| = \sqrt{1 - \rho(\lambda, \mu)^2}.$$

Choose $\omega \in \mathbb{T}$ so that $\langle u_\lambda, \omega u_\mu \rangle$ is nonnegative. Orthogonality of the image lines gives

$$2c^2 \leq \|A(u_\lambda - \omega u_\mu)\|^2 \leq \|A\|^2 \|u_\lambda - \omega u_\mu\|^2 = 2\|A\|^2 \big(1 - \sqrt{1 - \rho(\lambda, \mu)^2}\big).$$

This is (3). □

6 Audit of the 2024 candidate construction

Gamal's 2024 Theorem 6.3 [3] constructs finite-dimensional blocks

$$T_N = \begin{pmatrix} T_{N1} & \Gamma_N \\ 0 & T_{N0} \end{pmatrix} \approx T_{B_N}$$

such that $\sup_N M_{pb}(T_N) < \infty$ but $\sup_N M_{cpb}(T_N) = \infty$. The lower diagonal contains $d_N = 2^{N+1}$ pairwise distinct eigenvalues $\lambda_{N,l}$ in a fixed compact subdisk. A common disk automorphism is later applied to each block, and Theorem 7.1 combines the blocks into a cyclic polynomially bounded operator quasisimilar to T_B and not similar to a contraction. Corollary 2.3 then gives a polynomially bounded quasiaffine transform of S which is not similar to a contraction.

Theorem 5 (The direct 2024 extension misses the kernel estimate). *For the Theorem 6.3 block sequence used in the Theorem 7.1/Corollary 2.3 shift extension of [3], there is no quasiaffinity Z intertwining the resulting polynomially bounded operator $\mathbf{T}$ to S for which*

$$\inf_{\lambda \in \mathbb{D}} \|Z^* u_\lambda\| > 0.$$

Thus this construction does not answer the source question negatively.

Proof. Write

$$T_{N0} = \bigoplus_{n=0}^{N} D_N + \mathbf{S}_N,$$

where D_N is diagonal in an orthonormal basis $e_{N,1}, \ldots, e_{N,d_N}$. The chain subspaces generated by distinct $e_{N,l}$ are mutually orthogonal. Since the upper diagonal spectrum of T_N consists of distinct points $\nu_{N,j}$, none equal to any $\lambda_{N,l}$, the triangular form gives

$$\ker(T_N^* - \overline{\lambda_{N,l}} I) = \{0\} \oplus \ker(T_{N0}^* - \overline{\lambda_{N,l}} I).$$

Consequently these adjoint eigenlines are mutually orthogonal as l varies. The common disk automorphism used on block N preserves both these lines and all pseudohyperbolic distances.

Let $\zeta_{N,l}$ be the transformed eigenvalues. The construction may, and in the proof does, take the perturbation radii in a fixed compact subdisk (one may decrease each radius without affecting any estimate). Since $d_N \to \infty$, compactness of pseudohyperbolic balls implies

$$\min_{l \neq k} \rho(\zeta_{N,l}, \zeta_{N,k}) \longrightarrow 0 \tag{4}$$

along a subsequence.

The final $\mathbf{T}$ is polynomially bounded and $\mathbf{T} \prec S$. For every $\zeta \in \mathbb{D}$, the standard defect-one theorem for polynomially bounded quasiaffine transforms of S gives

$$\dim \ker(\mathbf{T}^* - \overline{\zeta} I) = 1$$

(apply the result at zero after a disk automorphism; see [1, 2]). At a zero $\zeta_{N,l}$ of B, the canonical Corollary 2.3 eigenvector has zero component in BH^2 and lies in the finite-block eigenline above. It is nonzero and hence spans the whole eigenspace. Therefore, for *any* intertwiner $Z\mathbf{T} = SZ$, $Z^* u_{\zeta_{N,l}}$ lies in that line. The image lines are mutually orthogonal for fixed N.

If the displayed uniform lower bound held, Lemma 4 applied to $A = Z^*$ would give one positive separation constant for every set $\{\zeta_{N,l} : 1 \leq l \leq d_N\}$. This contradicts (4). □

For the canonical Corollary 2.3 intertwiner Y, the same obstruction is visible in the exact identity

$$(1 - |\lambda|^2)\|Y^* h_\lambda\|^2 = |B(\lambda)|^2 + (1 - |\lambda|^2)\|Y_0^* h_{B,\lambda}\|^2.$$

At every zero of B, the shift summand disappears completely and the norm is forced onto the clustered finite-block eigenline. The outer product identity $X_0 Y_0 = g(T_B)$ from Theorem 7.1 does not repair this: its direct lower estimate contains $|g(\lambda)|$, and the chosen outer g tends rapidly to zero at the shifted blocks. A corona bound for (B, g) would instead make $g(T_B)$ invertible and force similarity of the base operator to the model.

7 Upgrade audit, limitations, and novelty status

Eight focused routes were pursued. Direct use of the 2024 examples led to the kernel identity above. The outer-product route failed at the necessary corona bound. Finite-block renormalization was ruled out by Lemma 4. Adding a coprime "good" model factor cannot help at an actual zero of B, because that eigenvector belongs to the B factor; sharing the zero raises geometric multiplicity and is incompatible with a quasiaffinity to a multiplicity-one shift. Merging each finite block into one very long cyclic Jordan chain preserves scalar polynomial estimates under sufficiently small finite-dimensional perturbations, but the intertwiner expands across the small connecting links and no uniform eigenvector normalization was obtained. The complete-polynomial-bound and weighted-shift routes give Theorems 1 and 2. Finally, bounded analytic interpolation gives Theorem 3, but there is no global interior-kernel sampling Riesz basis on which to close the argument.

A bounded novelty search used the exact source title and the phrases "polynomially bounded", "holomorphic eigenvectors", and "similar to the unilateral shift", together with later papers of the source author. The official arXiv records and TeX sources of arXiv:1901.03883 and arXiv:2412.14130 were inspected. The latter explicitly answers a different question of Kérchy but neither cites nor claims to resolve the 2019 eigenvector-norm question. No later explicit resolution of the exact question was found in this search.

Theorem 5 is an agent-derived implication from the published block construction, not a claim made in [3]. The principal human-review points are (i) the defect-one passage after disk automorphisms, (ii) the orthogonality of the lower diagonal eigenlines in Theorem 6.3, and (iii) the compact pseudohyperbolic packing step. The packet claims only the partial results above; the full polynomially bounded question remains open.

References

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